

Finite-size–corrected synchronization in a city-scale bus network: crossover, not criticality

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1. PROBLEM AND DATA

A bus that falls behind schedule meets more waiting passengers, dwells longer, and falls further behind, until the following bus catches it: the headways collapse and buses bunch. Newell and Potts framed this instability in 1964 [1], and a complex-systems reading treats each line as a ring of coupled self-oscillators whose phases lock under load, in analogy with the Kuramoto transition [2, 3]. That analogy predicts a demand-driven onset of collective synchronization. It has been examined almost entirely in simulation or on single shuttle loops; a city-scale, multi-line test with fused vehicle telemetry and passenger demand has been missing, largely for want of such data [4].

The NetMob26 dataset [5] supplies it for Niterói, Brazil. We use 19 days of telemetry (11–31 March 2026; 13,850,467 GPS fixes from 527 vehicles at 15–30 s spacing) and the full ticketing feed (6,790,681 boardings). The two feeds carry no common vehicle key, which fixes the identification scope of the study: passenger exposure is defined at the *line–direction–hour* level, and every demand-related quantity below is cell-level rather than vehicle-level. This is the unit at which supply and demand can be aligned in these data, and we treat it as the analysis unit by design.

2. SYNCHRONIZATION OBSERVABLE

For each of 20 high-ridership lines we project GPS onto the route polyline (`line_routes.json`, with `GTFS shapes.txt` as fallback), reduce position to arc length s , and assign a phase $\varphi = 2\pi s/L$ per direction, so the buses on a (*line, direction, hour*) cell become oscillators on a ring. After dropping routes that fail map-matching and excluding terminal-layover and sparsely sampled bus-hours (69,466 active bus-hours retained), we compute the first Kuramoto–Daido order parameter

$$r_1 = \left| \frac{1}{N} \sum_{j=1}^N e^{i\varphi_j} \right|, \quad (1)$$

near 0 for evenly spread buses (the splay state) and near 1 when they bunch.

The order parameter is not comparable across cells. For N uniformly random phases $\mathbb{E}[r_1] \approx 0.886/\sqrt{N}$, so a cell with few buses reports a high r_1 with no underlying coordination, and any quantity that covaries with fleet size — weekends, off-peak hours, low-demand lines — inherits a spurious synchronization signal. We therefore draw 5,000 random-phase samples at each observed N and report a finite-size–corrected synchronization,

$$r_{\text{exc}} = \frac{r_1 - \mathbb{E}_{\text{null}}[r_1 | N]}{1 - \mathbb{E}_{\text{null}}[r_1 | N]}, \quad z = \frac{r_1 - \mathbb{E}_{\text{null}}[r_1 | N]}{\text{sd}_{\text{null}}[r_1 | N]}, \quad (2)$$

which measure how far a cell sits above its same- N random baseline. This correction is the central methodological contribution of the paper: in a transit system with fluctuating fleet sizes it is a prerequisite for any field measurement of synchronization, and every value below is reported with its N .

3. RESULTS

Result 1 — finite-size correction reverses the apparent weekend effect. In the raw signal weekends look *more* synchronized than weekdays (weekend–weekday $r_1 = +0.047$, 95% CI [0.039, 0.056]), despite running far fewer buses. This is exactly the regime where r_1 is least trustworthy, and the correction overturns it: weekend–weekday $r_{\text{exc}} = -0.067$ ([−0.085, −0.048]) and $z = -0.441$ ([−0.486, −0.397]); corrected synchronization is *lower* on weekends (Table 1). The reversal is not a cosmetic adjustment — it changes the substantive conclusion about when the system is most coordinated — and it is the clearest empirical demonstration that raw Kuramoto observables cannot be compared across cells of different N .

Result 2 — demand modulates synchronization, but without a sharp transition. We summarize cell-level load as $\lambda = \text{boardings}/N$. Because N is itself an operational response to demand and conditions, λ should be read as realized load per active bus, not exogenous passenger demand. Corrected synchronization rises with λ (standardized random-intercept coefficient 0.039, $p \approx 3 \times 10^{-15}$), and a piecewise fit places an apparent break at $\lambda \approx 136$ boardings/bus. We treat this as an observational crossover scale, not a critical point, because a pre-registered three-part gate does not support criticality. The susceptibility check asks whether fluctuations in r_{exc} peak at an interior λ (the signature of a transition); within N -strata they do not (peak/edge ratios 0.89–1.14). The coexistence check asks whether low- and high-synchronization states co-occur near the candidate scale; the order-parameter distribution is no more bimodal there than elsewhere ($\Delta\text{BIC} \approx 18$, weaker than off-scale regimes). The N -stability check asks whether the apparent scale is more than fleet-size composition; this one passes. With one of three signatures met, the honest reading is a smooth, demand-modulated crossover rather than city-scale criticality.

Result 3 — mechanisms are operational and common-shock dominated. We examined two routes by which a genuine oscillator-coupling story might persist, and both resolve toward operational explanations. *Downstream amplification.* Headway irregularity grows along the route interior (mean headway-CV slope 0.12, $p \approx 2 \times 10^{-58}$ on the 10–90% interior), the spatial fingerprint of the Newell–Potts instability. In a standardized model (Fig. 2) this growth is more strongly associated with the irregularity already inherited upstream (−0.283, [−0.296, −0.270]) than with contemporaneous realized load (0.015, [0.002, 0.027]) or fleet size (0.081), with route topology unresolved; the negative sign on the upstream term reflects a ceiling — already-irregular hours have less room to grow downstream — not a stabilizing effect of irregularity. *Dwell mediation.* The classic $\lambda \rightarrow$ dwell-time variability $\rightarrow$ bunching channel is not established in these data (indirect effect ≈ 0.0001 , proportion mediated 0.2%), partly because boardings are not linked to individual vehicles and GPS-derived dwell is noisy; we do not read this as ruling the mechanism out. *Shared corridors.* Lines on a common trunk are more co-synchronized on the shared segment than off it (within-pair raw gap 0.055,

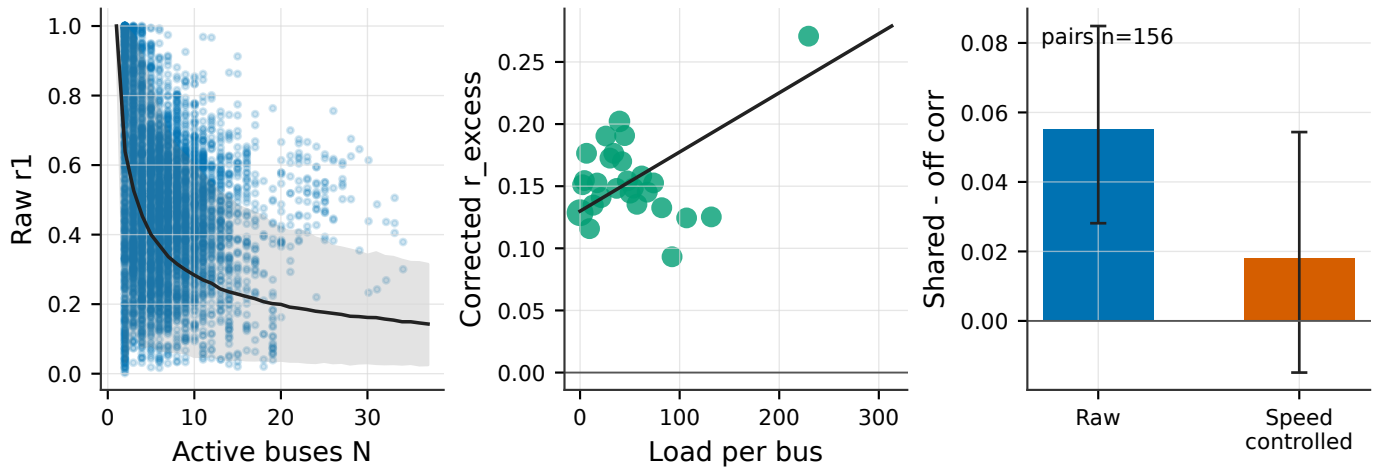


Fig. 1. The argument in one figure. Raw r_1 is finite-size confounded, tracking the active fleet size rather than coordination; correcting against a same- N random-phase null reverses the apparent weekend effect (weekends are *less* synchronized once corrected); the dependence of corrected synchronization on realized load per bus λ is a smooth crossover, with no sharp onset; and the cross-line signal on shared corridors attenuates toward zero once the common road state (segment speed) is controlled. Panels reproduced from the Phase 4 synthesis output.

Table 1. Corrected synchronization by day type and period^a

Group	N	r_1	r_{exc}	z	λ
Weekday	7.99	0.482	0.179	0.806	57.6
Weekend	4.47	0.529	0.112	0.366	40.3
Weekday, off-peak	7.23	0.499	0.178	0.758	63.7
Weekday, peak	10.82	0.420	0.180	0.985	34.7
Weekend, off-peak	4.34	0.537	0.113	0.356	41.2
Weekend, peak	4.95	0.504	0.109	0.400	37.3

^aRaw r_1 is higher on weekends; finite-size-corrected r_{exc} and z are lower. λ is realized boardings per active bus.

[0.028, 0.085]; 156 pairs). Controlling for the shared-segment mean bus speed — the common road state — the gap falls to 0.018 ([-0.015, 0.054]), retaining about a third and now spanning zero, and lead-lag and transfer-entropy tests show no propagation (5.8% of pairs with a non-zero-lag peak; net directional asymmetry $p = 0.87$; Fig. 1, right panel). The co-synchronization is largely consistent with shared road-state exposure; a residual inter-line coupling is not established, though speed control may also absorb part of a genuine coupling channel.

4. CONCLUSION

What appears as collective synchronization in raw transit data can be a mixture of finite-size bias, operational dispatch irregularity, and common road-state shocks. In Niterói, once the order parameter is corrected against its same- N random baseline, the demand dependence is a smooth crossover rather than a critical transition; downstream amplification tracks inherited irregularity more than realized load; and shared-corridor co-synchronization is largely a shared road state. The methodological point generalizes: empirical synchronization observables must be corrected against finite-size null models before they are interpreted as collective dynamics or critical phenomena, in transit and in other finite, fluctuating ensembles measured in

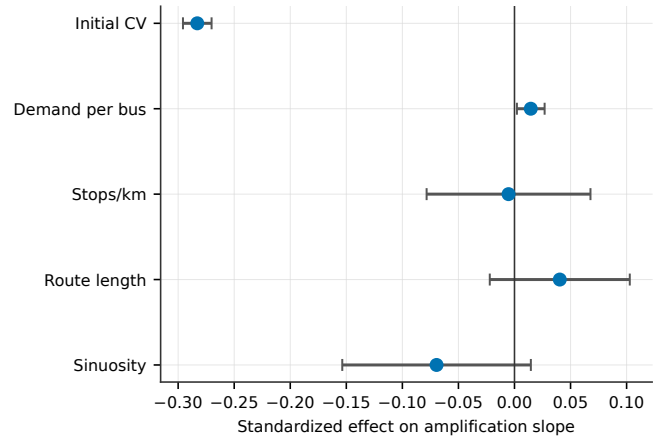


Fig. 2. Inherited upstream irregularity, not realized load, is the strongest predictor of downstream headway-CV growth. Standardized effects from a line-direction random-intercept model with 95% CIs; the negative upstream-irregularity coefficient is a ceiling effect. Realized load per bus is small and route topology is unresolved.

the field. Two steps would sharpen identification further — a propagation-aware coupling model able to separate inter-line transfer from shared road shocks, and vehicle-linked boarding data with a cleaner dwell signal to test the demand mechanism directly.

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